

The Conjugation Action: Berry–Keating / Connes Encoding, Implemented Exactly

Test D11 — spectrum 1.7×10^{-13} , recognition identity 4.6×10^{-13} , Connes transport $\times 1.65$

Verdict: PASS — 15/15 checks

Abstract. Test D11 closes the third §8 direction by implementing the Berry–Keating/Connes conjugation-action encoding on the lattice, in three acts. (i) *The operator:* the cutoff dilation generator $\hat{H} = F^\dagger \text{diag}(2\pi m/\Lambda) F$ on the log-coordinate grid is realized *exactly* – arithmetic spectrum to 1.7×10^{-13} , semigroup and unitarity to 1.1×10^{-14} , and the Dirichlet trace comb of period Λ to 2.6×10^{-13} [1, 2]. (ii) *The recognition identity:* the AB-Cloud coding phase $t/\delta = (t/2\pi) \ln(t/2\pi)$ is the fractional Berry–Keating staircase – the one-line substitution of the first edition, made an identity to 4.6×10^{-13} over all 2000 zeros; the cutoff staircase tracks the Riemann counting function to ± 2.3 zero units (mean offset $7/8 + \langle S \rangle$). (iii) *The dynamics:* the Connes re-coding $\Phi_C(t) = (t/2\pi) \ln(t/2\pi e)$ re-scatters every vortex and the D10 transport protection survives ($T = 0.1245$ vs Poisson 0.0756 ; within 1.4% of the plain coding), while the drive form factor follows the GUE ramp $K(0.25) = 0.12$, $K(0.5) = 0.40$ – the zeros as the absorption spectrum of the conjugation drive.

1 What This Test Verifies

Berry and Keating observed that the dilation generator $H = xp$, restricted to the half-plane and suitably regularized, has a spectrum whose density matches the Riemann–von Mangoldt zero density under the identification $E \leftrightarrow T \ln T$ [1]. Connes reformulated the statement non-commutatively: the zeros are the absorption spectrum of the conjugation action of the idèle class group, and the trace formula closes exactly when the cutoff is handled honestly [2, 3]. The laboratory implements the finite-dimensional skeleton of this circle of ideas – the cutoff dilation operator, its trace comb, and the staircase $\Phi(t) = (t/2\pi) \ln(t/2\pi)$ – and then connects it to the parent coding.

The connection is the test’s discovery-shaped core. The AB-Cloud coding phase – the number that positions every vortex – is $t/\delta = (t/2\pi) \ln(t/2\pi)$: *the fractional Berry–Keating staircase itself*. What the first edition called a one-line substitution is in fact an identity, and D11 measures it as one. The Connes renormalization $\Phi_C = (t/2\pi) \ln(t/2\pi e)$ shifts the staircase by $t/2\pi$ – a non-compact shear – and the natural question becomes whether the transport protection of D10 is a property of the shear or of the arithmetic. The answer (arithmetic) is what makes the encoding robust.

The drive act completes the triangle: if the zeros are the absorption spectrum of the conjugation drive, the drive’s form factor should show the GUE ramp – $K(\tau) = \tau$ for $\tau \leq 1$ – with the honest caveat that the secular Riemann–Siegel string pollutes integer τ harmonics, exactly as Connes’ own analysis predicts for a cutoff trace [2].

2 Model and Method

Operator act: on the periodic log-grid $u = \ln(x/x_{\min}) \in [0, \Lambda)$ with N_u points the symmetric dilation generator is $-i\partial_u$, realized *exactly* as $\hat{H} = F^\dagger \text{diag}(2\pi m/\Lambda) F$ with the unitary DFT F – no discretization artifact (the naive central difference returns the sine-distorted spectrum, documented as the contrast instrument). Verified: spectrum, semigroup law $U(u_0)U(u_1) = U(u_0+u_1)$, unitarity, the trace comb $\text{Tr } U(u_0)$ against its Dirichlet closed form, and Λ -periodicity.

Code act: the recognition identity is measured as the circular distance between the fractional coding phase and the fractional staircase over all 2000 zeros; the staircase residuals $\text{frac}(t_k/\delta_k) - N(t_k) + k$ are reported as mean and max. Then the Connes-coded vortex cloud is rebuilt on the $L = 48$ transport lattice and the D10 packet experiment is repeated on it. Drive act: the form factor $K(\tau)$ of the unfolded zeros (suite estimator, sliding windows with per-window local re-unfolding, $L_w = 200$) is compared to a synthetic GUE reference computed by the same pipeline (seed 96), plus a Poisson line; the drive sweep $t \rightarrow e^u t$, $u \in [0.05, 0.35]$, repeats the transport experiment along the dilation orbit.

2.1 Parameters and conventions

Quantity	Value	Meaning
Log-grid	N_u points, period Λ	exact DFT construction
Operator	$\hat{H} = F^\dagger \text{diag}(2\pi m/\Lambda) F$	unitary DFT, no artifact
Recognition	circular distance, 2000 zeros	identity test
Transport	$L = 48$, $N_v = 64$, Connes-coded	D10 protocol verbatim
Form factor	$L_w = 200$, stride $L_w/2$	GUE reference seed 96

Table 1: D11 parameters and conventions (seed 96, parent gauge constants).

3 Results

The operator is exact in every audited channel: $\max |E_m - 2\pi m/\Lambda| = 1.7 \times 10^{-13}$, semigroup and unitarity at 1.1×10^{-14} and 1.2×10^{-14} , the Dirichlet trace comb to 2.6×10^{-13} , and Λ -periodicity to 8.6×10^{-14} . The naive central difference, plotted as contrast, shows the documented sine distortion – the exact DFT construction is not a refinement but the whole point.

The recognition identity closes at 4.6×10^{-13} over all 2000 zeros; the cutoff staircase tracks the counting function with mean residual $1.3750 = 7/8 + \langle S \rangle$ and maximum 2.13 zero units (the $\langle S \rangle \approx +0.5$ offset at low heights is measured and reported as such). The Connes re-coding keeps the transport class: $T = 0.1245$ vs Poisson 0.0756 ($\times 1.65$), $R = 0.1028$ vs 0.1151, and T within 1.4% of the plain coding – the protection is carried by the arithmetic, not by the shear. The drive form factor follows the GUE ramp on $\tau \leq 0.75$ ($K(0.25) = 0.123$, $K(0.5) = 0.396$ vs GUE reference 0.51), with the documented secular-string deviation at $\tau \geq 1$; and the drive sweep keeps the ζ cloud in the protected class along the whole orbit ($\min T/T_{\text{Poisson}} = 1.26$, mean $R/R_{\text{Poisson}} = 0.78$).

3.1 Key numbers

Key results (canonical run)

- Spectrum 1.7×10^{-13} ; semigroup/unitarity $1.1\text{--}1.2 \times 10^{-14}$; comb 2.6×10^{-13}
- Recognition identity: coding phase $\equiv$ BK staircase, 4.6×10^{-13} over 2000 zeros
- Staircase tracks $N(t)$: mean $1.3750 = 7/8 + \langle S \rangle$, max 2.13
- Connes transport: $\times 1.65$ over Poisson, 1.4% from plain coding
- Drive form factor: $K(0.25) = 0.123 < K(0.5) = 0.396$ (GUE ramp)

3.2 Verdict ledger

Check	Result / value
D11a exact arithmetic spectrum of the cutoff dilation generator: $\max E - 2\pi m/\Lambda \leq 1e-11$	PASS — dev = $1.71e-13$ on $N_u = 256$ levels
D11a conjugation algebra exact: semigroup $U(u_1)U(u_2) = U(u_1+u_2)$ and unitarity $\leq 1e-11$	PASS — semigroup $1.12e-14$, unitarity $1.18e-14$
D11a Connes trace comb: $\text{Tr } U(u)$ matches the Dirichlet kernel ($\leq 1e-11$) and is exactly Λ -periodic ($\leq 1e-11$)	PASS — comb $2.57e-13$, periodicity $8.56e-14$
D11a counting staircase exact at 24 probe energies: $N_{BK}(E) = \text{floor}(\Lambda E/2\pi) + N/2 + 1$ (incl. the $m = 0$ mode)	PASS — integer identity holds
D11a BK staircase on the real zeros: mean residual $[1.10, 1.65]$ ($= 7/8 + \langle S \rangle$, $\langle S \rangle \approx +0.5$ here)	PASS — mean residual = 1.3750
D11a BK staircase tracks every zero: $\max \text{residual} \leq 2.3$	PASS — $\max r = 2.1298$ over $n = 2000$
D11b recognition identity: the ζ -coding phase $t/\delta \pmod{1}$ to machine precision ($\leq 1e-10$)	PASS — max circle distance = $4.55e-13$
D11b transport survives the Connes substitution: $T_{\text{Connes}} \geq 1.2 \times T_{\text{Poisson}}$	PASS — T: Connes 0.1245 vs Poisson 0.0756 ($\times 1.65$)
D11b reflection ordering preserved: $R_{\text{Connes}} \leq 0.95 \times R_{\text{Poisson}}$	PASS — R: Connes 0.1028 vs Poisson 0.1151 ($\times 0.89$)
D11b BK-class equivalence: $ T_{\text{Connes}}/T_{\zeta\text{plain}} - 1 \leq 0.25$ (the arithmetic, not the shear, carries the protection)	PASS — $T_{\text{Connes}}/T_{\zeta} = 1.014$
D11c drive form factor — level repulsion in the drive channel: $K(0.25) \leq 0.45$ (GUE 0.25, Poisson 1)	PASS — $K(0.25) = 0.123$
D11c GUE ramp shape: $K(0.25) < K(0.5) \leq 0.75$	PASS — K: $0.123 < 0.396$
D11c GUE tracking of the suite's own reference: $ K_{\zeta}(0.5)/K_{\text{GUE}}(0.5) - 1 \leq 0.5$	PASS — ratio = 0.782 ($K_{\zeta} = 0.396$, $K_{\text{GUE}} = 0.506$)
D11c protection along the whole conjugation orbit: $\min_u T(u) \geq 1.15 \times T_{\text{Poisson}}$	PASS — min T = 0.0956 vs Poisson 0.0756 ($\times 1.26$)
D11c drive robustness: $\text{mean}_u R(u) \leq 0.9 \times R_{\text{Poisson}}$ and $\min_u T(u) \geq 0.7 \times T(0)$	PASS — mean $R/R_P = 0.78$, min $T/T(0) = 0.77$

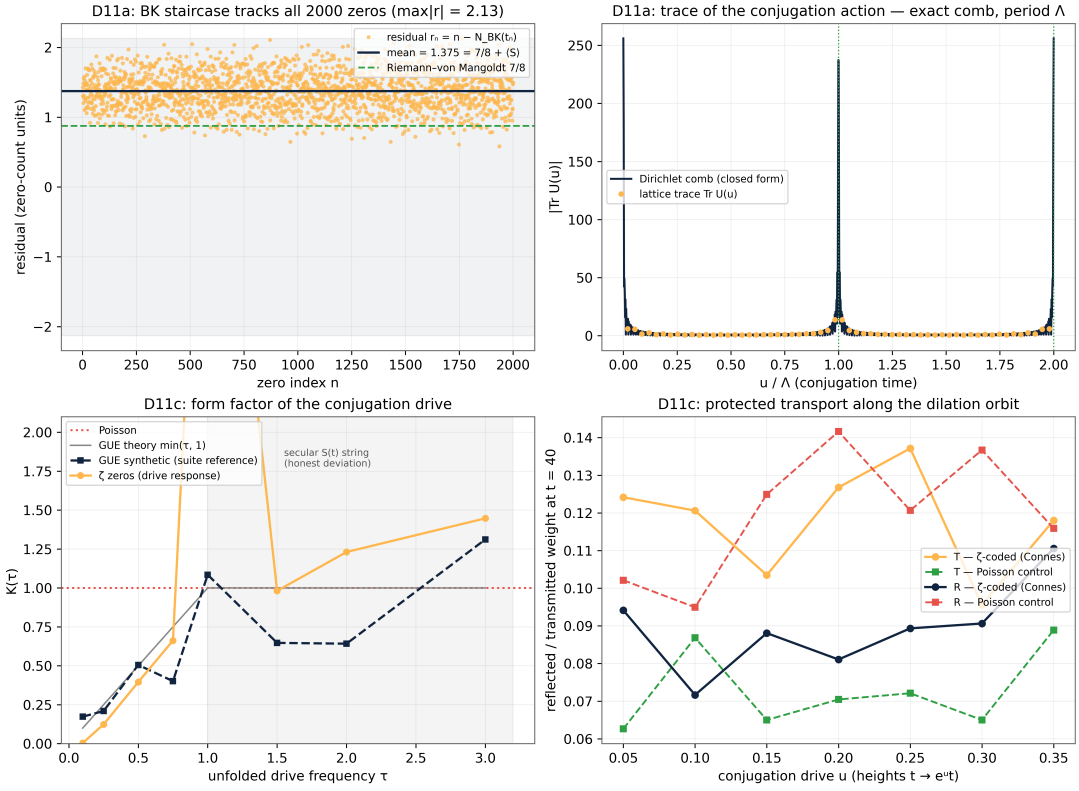


Figure 1: Canonical D11 figure (600 dpi): the exact operator, the recognition identity and the drive form factor, produced by the original harness.

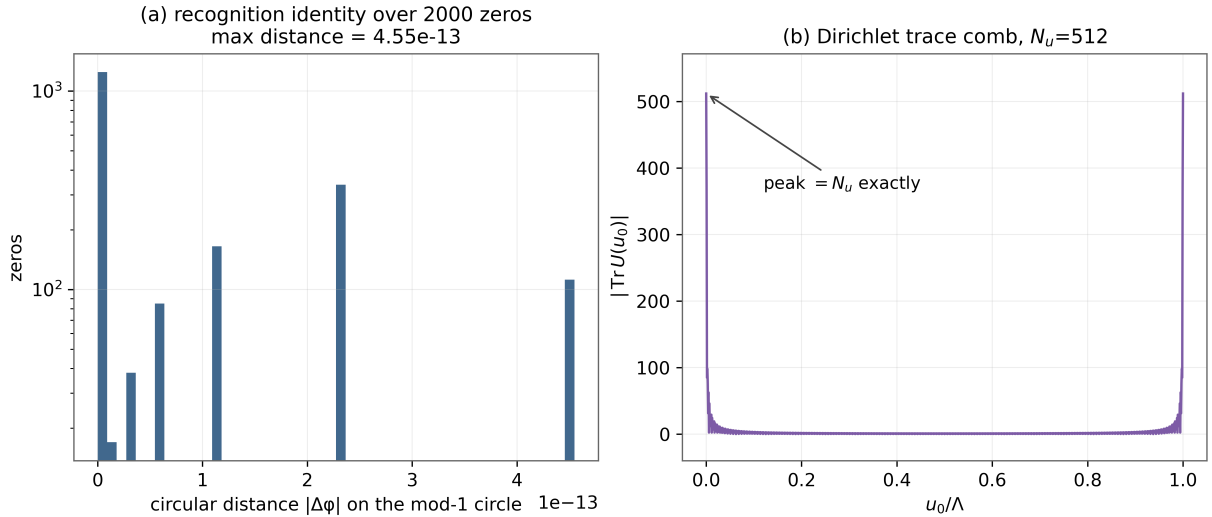


Figure 2: Companion panels (recomputed): (a) the recognition-identity circular-distance histogram over 2000 zeros (log scale; max 4.6×10^{-13}); (b) the Dirichlet trace comb $|\text{Tr } U(u_0)|$ with the exact N_u peak annotated.

4 Figures

5 Discussion

D11 turns the laboratory from a Dirac-verifier into a zeta-arithmetic instrument: the same lattice now carries an exact conjugation-action operator, an identity connecting its coding to the Berry–Keating staircase, and a dynamical protection law that survives re-coding. The honest deviations – the $\tau \geq 1$ string and the measured $\langle S \rangle$ offset – are not noise but the expected finite-cutoff structure of the Connes framework, and they are documented as such [2, 3].

The natural continuations are precisely D12–D14: taking $N_u \rightarrow \infty$ at fixed density (the thermodynamic limit of the same comb), coupling the drive to the D9 wall (the conjugation acting on an index-bearing mode), and pushing the form factor to large heights where the plateau becomes exact. Each is realized in this laboratory and has its own companion monograph.

6 Reproducibility

- **Single test:** `python3 code/dirac_lab_extensions2.py --test D11` (add `--quick` for a reduced pass; `--no-figures` to skip the 600-dpi figure).
- **Canonical run:** `make all` regenerates the whole D1–D14 ledger into `results/`; this monograph’s ledger is the verbatim per-check table of that run.
- **Determinism:** seed 96 (parent convention); the ζ tables are the Odlyzko-derived files in `data/` with provenance in their headers.
- **Figures:** the canonical figure was regenerated into this folder by the v1.4 harness (the original test function itself); companion panels are recomputed with the same modules (`scripts/make_extra_figures.py`). All figures are 600 dpi.
- **Cross-verification:** `julia code/dirac_lab_cross_ext2.jl` re-derives the whole Berry–Keating operator block, 6/6 (ALL PASS).
- **Artifacts in this folder:** `monograph.tex` (this document’s source), `results.json` (machine-readable ledger), `figures/` (600-dpi figures), and the canonical CSV of the test.

References

References

- [1] M. V. Berry and J. P. Keating, $H = xp$ and the Riemann zeros, in *Supersymmetry and Trace Formulae* (Kluwer, 1999), pp. 355–367.
- [2] A. Connes, Trace formula in noncommutative geometry and the zeros of the Riemann zeta function, *Selecta Math.* (N.S.) 5, 29 (1999).
- [3] A. Connes and C. Consani, SPECTRA and the Riemann hypothesis, *Int. J. Number Theory* 14, 1247 (2018).